

OPNsense Firewall Policy Validation Lab

Firewall Configuration, Traffic Validation, Logging and Verification

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Introduction

This report presents the OPNsense Firewall Policy Validation Lab performed in a virtual lab environment using OPNsense, Kali Linux, Ubuntu Server, and Windows. During the lab, firewall rules were configured on the Homelab interface to control network traffic based on different protocols and services.

The configured firewall policies were verified by generating traffic from Kali Linux using tools such as ping, nmap, nslookup, ssh, and a web browser. Both allowed and blocked traffic were tested, and the results were verified through firewall logs and connectivity tests.

This report documents the firewall rule configuration, traffic validation, rule ordering, logging, and verification process exactly as performed during the lab, with each step supported by the corresponding screenshots.

Configuring the Allow HTTP Rule

The first rule created on the Homelab interface allows outbound HTTP traffic on TCP port 80. The Protocol field is set to TCP, the Source is set to Homelab network, and the Destination is left as any so that any host on the Homelab network can reach web servers on port 80.

The Destination port range is set to HTTP (80) for both the from and to fields, and the rule description is entered as "Allow http" so the rule is easy to identify later in the rule list.

The screenshot shows the OPNsense Firewall Rule configuration interface. The rule is named "Allow http". The configuration is as follows:

- Protocol:** TCP
- Source / Invert:** ☐ Use this option to invert the sense of the match.
- Source:** Homelab network
- Source port range:** from: any, to: any
- Destination / Invert:** ☐ Use this option to invert the sense of the match.
- Destination:** any
- Destination port range:** from: HTTP (80), to: HTTP (80)
- Log:** ☐ Log packets that are handled by this rule
- Category:** (empty)
- Description:** Allow http

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Figure 1: Allow HTTP Rule Configuration on the Homelab Interface

Configuring the Allow HTTPS Rule

The next rule is created for HTTPS traffic under Firewall → Rules → Homelab. The Protocol is set to TCP, Source is Homelab network, and Destination is any, matching the pattern used for the HTTP rule.

The Destination port range is set to HTTPS (443), and the Description field reads "Allow https". This rule allows secure web browsing traffic to pass out from the Homelab network.

The screenshot shows the OPNsense Firewall Rule configuration interface for the "Allow https" rule. The rule is named "Allow https". The configuration is as follows:

- Protocol:** TCP
- Source / Invert:** ☐ Use this option to invert the sense of the match.
- Source:** Homelab network
- Destination / Invert:** ☐ Use this option to invert the sense of the match.
- Destination:** any
- Destination port range:** from: HTTPS (443), to: HTTPS (443)
- Log:** ☐ Log packets that are handled by this rule
- Category:** (empty)
- Description:** Allow https
- No XMLRPC Sync:** ☐
- Schedule:** none

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Figure 2: Allow HTTPS Rule Configuration on the Homelab Interface

Configuring the Allow DNS Rule

A DNS rule is added so that hosts on the Homelab network can resolve domain names. The Destination port range is set to DOMAIN (53), which covers DNS lookups, and the Description is set to "Allow DNS".

The Source remains Homelab network, Destination is any, Schedule is none, and the Gateway is left at default. With HTTP, HTTPS, and DNS now permitted, the next rules focus on blocking insecure services.

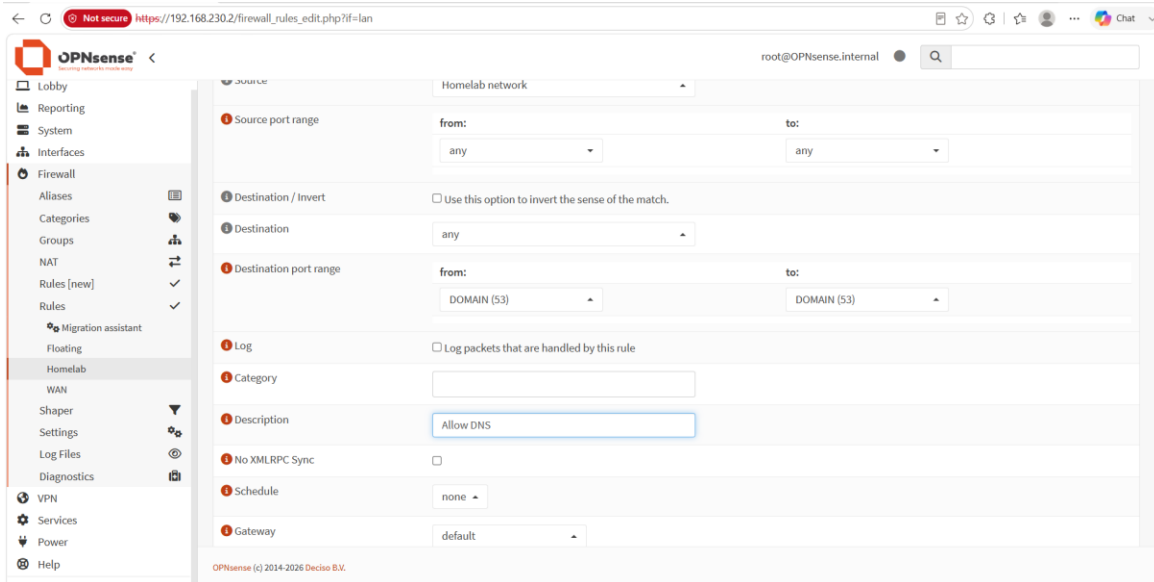


Figure 3: Allow DNS Rule Configuration on the Homelab Interface

Creating the Block Telnet Rule

To prevent the use of the insecure Telnet protocol, a block rule is created with the Destination port range set to TELNET (23). The Description is entered as "Block telnet" so the rule's purpose is clear in the rule list.

This rule sits on the Homelab interface alongside the earlier allow rules and is used later to confirm that Telnet connections from Kali Linux are refused.

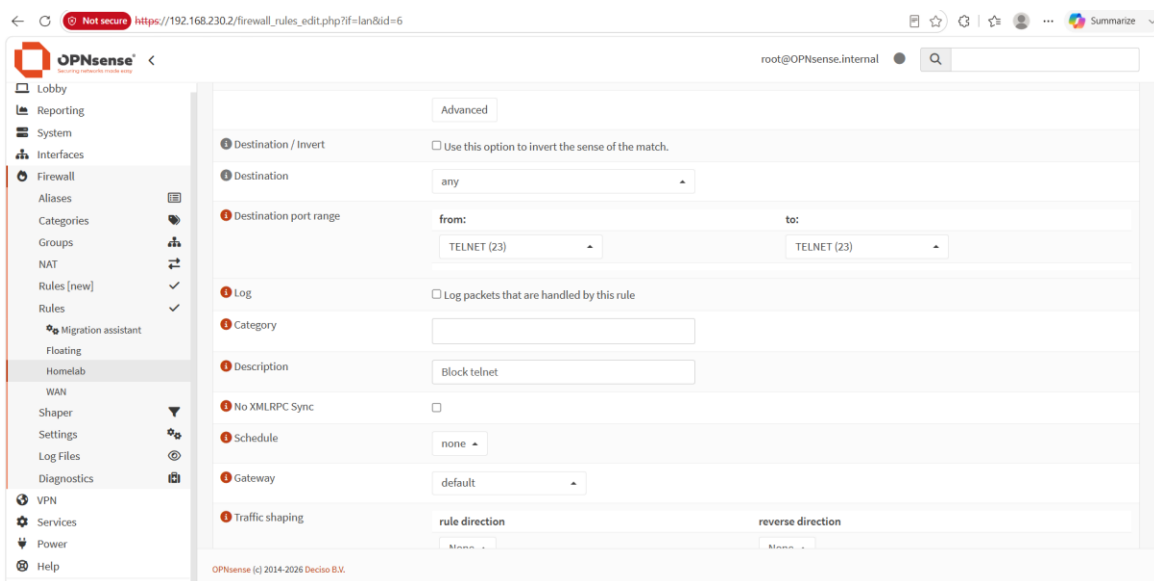


Figure 4: Block Telnet Rule Configuration on the Homelab Interface

Creating the Block SMB Rule

A similar block rule is created for SMB file-sharing traffic. The Destination port range is set to MICROSOFT-DS (445), and the Description reads "Block SMB".

Together with the Telnet block rule, this stops two commonly abused protocols from being reachable on the Homelab network while HTTP, HTTPS, DNS, and SSH remain permitted.

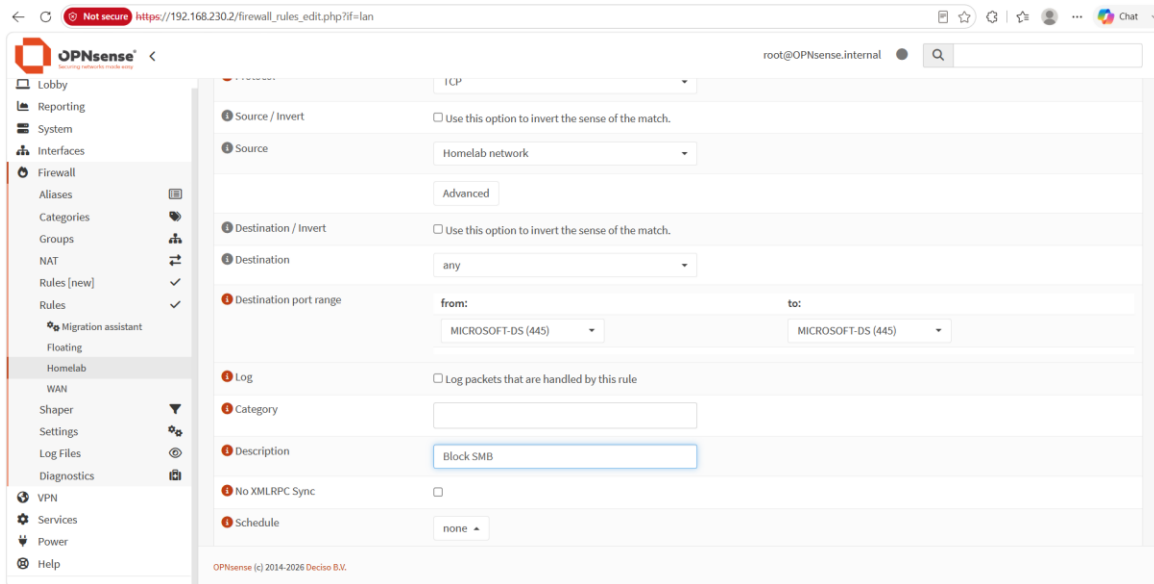


Figure 5: Block SMB Rule Configuration on the Homelab Interface

Configuring the Allow SSH Rule

An SSH access rule is created so that the Ubuntu server can still be managed remotely. The Interface is Homelab, Direction is in, TCP/IP Version is IPv4, and Protocol is TCP.

The Source is Homelab network, and the Destination is set to Single host or Network with the value 192.168.230.177, which is the Ubuntu server's address. The Destination port range is set to SSH (22), restricting SSH access specifically to that host instead of the whole network.

1 Interface	Homelab
2 Direction	in
3 TCP/IP Version	IPv4
4 Protocol	TCP
5 Source / Invert	☐ Use this option to invert the sense of the match.
6 Source	Homelab network
Advanced	
7 Destination / Invert	☐ Use this option to invert the sense of the match.
8 Destination	Single host or Network 192.168.230.177
9 Destination port range	from: SSH (22) to: SSH (22)
10 Log	☐ Log packets that are handled by this rule

Figure 6: Allow SSH Rule Configuration for the Ubuntu Server

Creating the Block ICMP Rule

The final new rule blocks ICMP traffic directed at the Ubuntu server. Quick is enabled so the action is applied immediately on match, the Interface is Homelab, Direction is in, and TCP/IP Version is IPv4.

The Protocol is set to ICMP with ICMP type Any, Source is Homelab network, and Destination is Single host or Network set to 192.168.230.177. This rule is used later to demonstrate that ping requests to the Ubuntu host are dropped.

1 Quick	☒ Apply the action immediately on match.
2 Interface	Homelab
3 Direction	in
4 TCP/IP Version	IPv4
5 Protocol	ICMP
6 ICMP type	Any
7 Source / Invert	☐ Use this option to invert the sense of the match.
8 Source	Homelab network
Advanced	
9 Destination / Invert	☐ Use this option to invert the sense of the match.
10 Destination	Single host or Network 192.168.230.177
11 Destination port range	from: to:

Figure 7: Block ICMP Rule Configuration for the Ubuntu Server

Reviewing the Complete Rule Set on the Homelab Interface

After all seven rules are created, the Homelab rule list is opened to confirm everything applied correctly. The banner "The changes have been applied successfully" confirms the configuration was saved without errors.

The list shows the default allow LAN rules at the top, followed by Allow http, Allow https, Allow DNS, and allow ssh (scoped to 192.168.230.177), and finally the Block telnet, Block SMB, and Block ICMP rules marked with a red X for block. Because OPNsense evaluates rules from top to bottom on a first-match basis, having the default allow rules ahead of the block rules means the block rules would not yet take effect during testing, which is examined further later in the report.

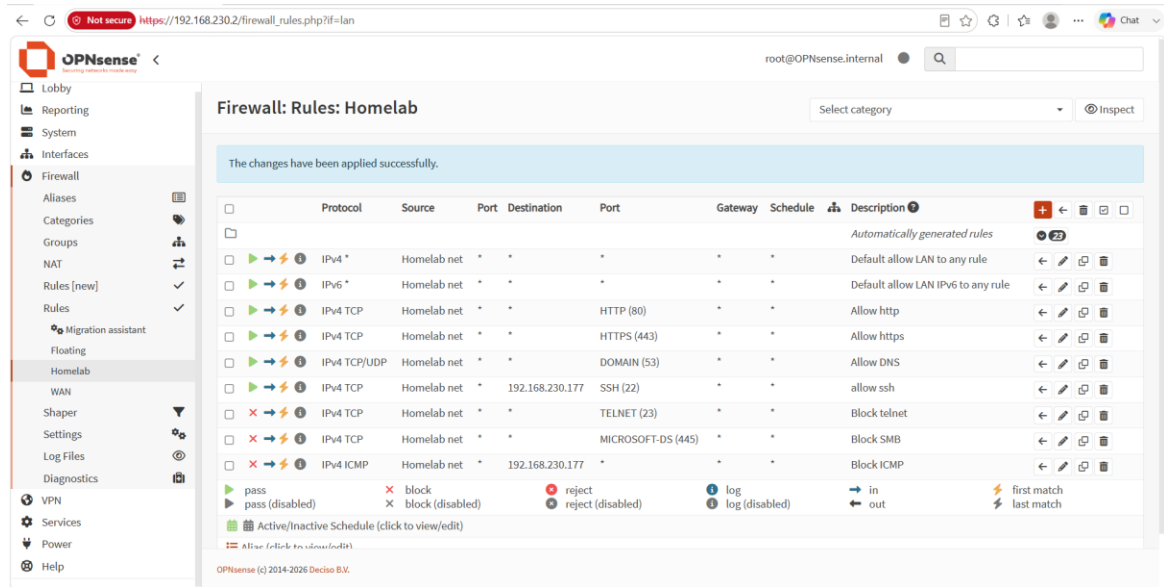


Figure 8: Final Rule Set on the Homelab Interface with the Default Allow Rule at the Top

Testing Allowed HTTP Traffic from Kali Linux

With the rules in place, HTTP connectivity is tested from the Kali Linux virtual machine using Firefox. The browser successfully loads the NeverSSL test page over plain HTTP, confirming that port 80 traffic from the Homelab network reaches its destination.

This result verifies that the Allow HTTP rule is functioning as intended before moving on to test DNS resolution.

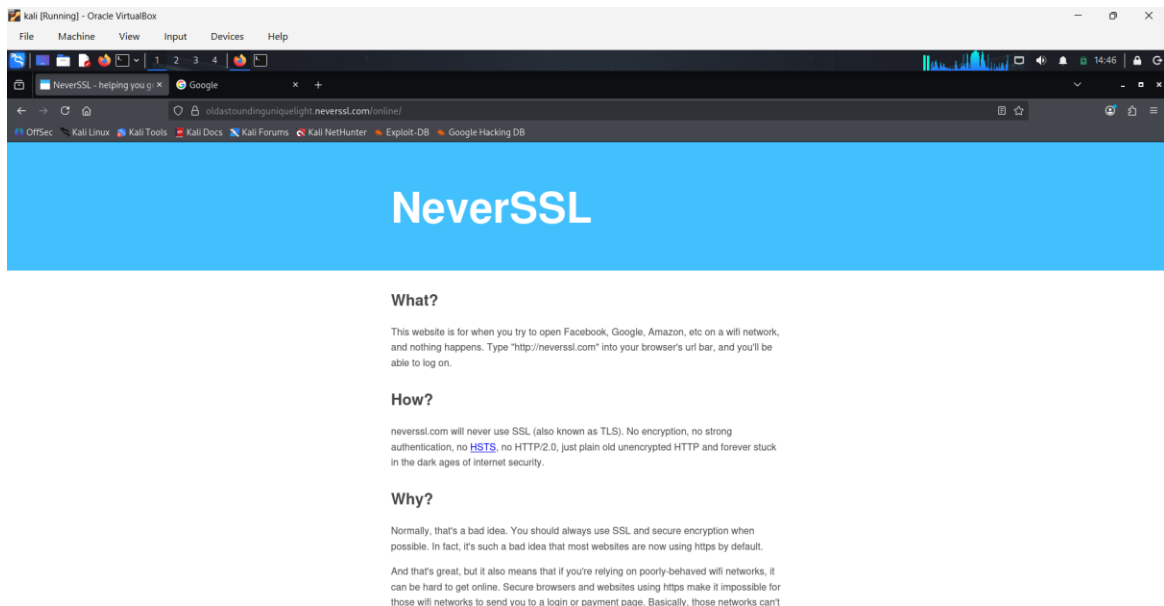


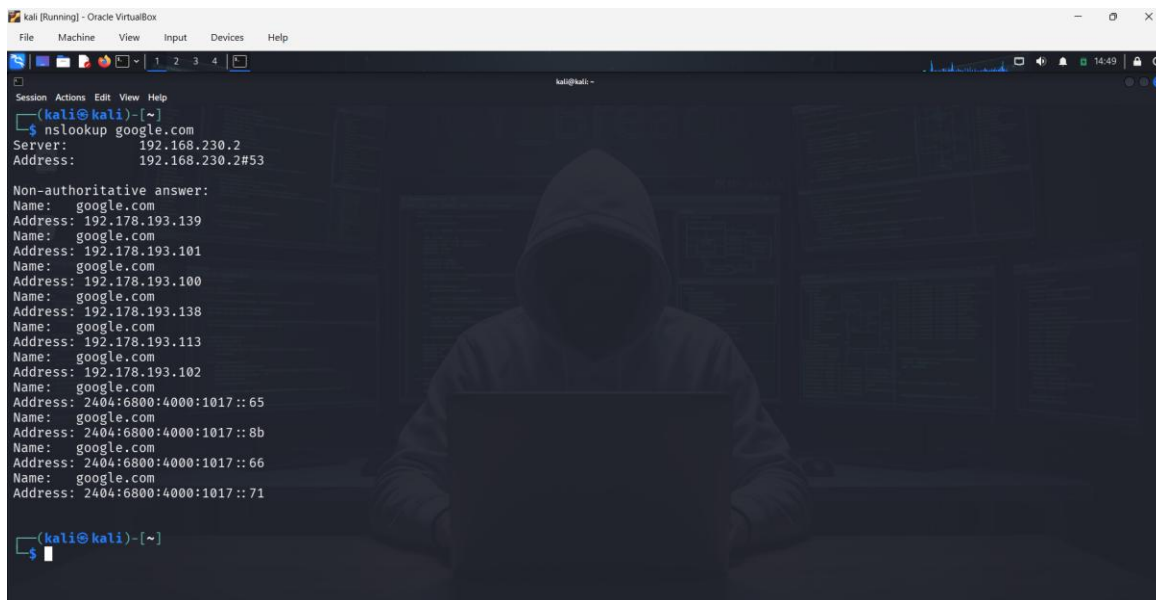
Figure 9: Successful HTTP Page Load from Kali Linux Confirming Port 80 Is Allowed

Verifying DNS Resolution from Kali Linux

DNS resolution is tested by running the following command in the Kali Linux terminal:

```
nslookup google.com
```

The output shows the query being answered by the server at 192.168.230.2 on port 53, which is the OPNsense firewall, and returns multiple IPv4 and IPv6 addresses for google.com. This confirms the Allow DNS rule is working correctly.



```
kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

(kali@kali)~$ nslookup google.com
Server:      192.168.230.2
Address:     192.168.230.2#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.193.139
Name:   google.com
Address: 192.178.193.101
Name:   google.com
Address: 192.178.193.100
Name:   google.com
Address: 192.178.193.138
Name:   google.com
Address: 192.178.193.113
Name:   google.com
Address: 192.178.193.102
Name:   google.com
Address: 2404:6800:4000:1017::65
Name:   google.com
Address: 2404:6800:4000:1017::8b
Name:   google.com
Address: 2404:6800:4000:1017::66
Name:   google.com
Address: 2404:6800:4000:1017::71

(kali@kali)~$
```

Figure 10: DNS Resolution Verification Using nslookup from Kali Linux

Verifying HTTPS Access from Kali Linux

HTTPS access is checked by browsing to www.google.com from the Kali Linux Firefox browser. The page loads successfully over a secure connection, confirming that the Allow HTTPS rule permits port 443 traffic out from the Homelab network.

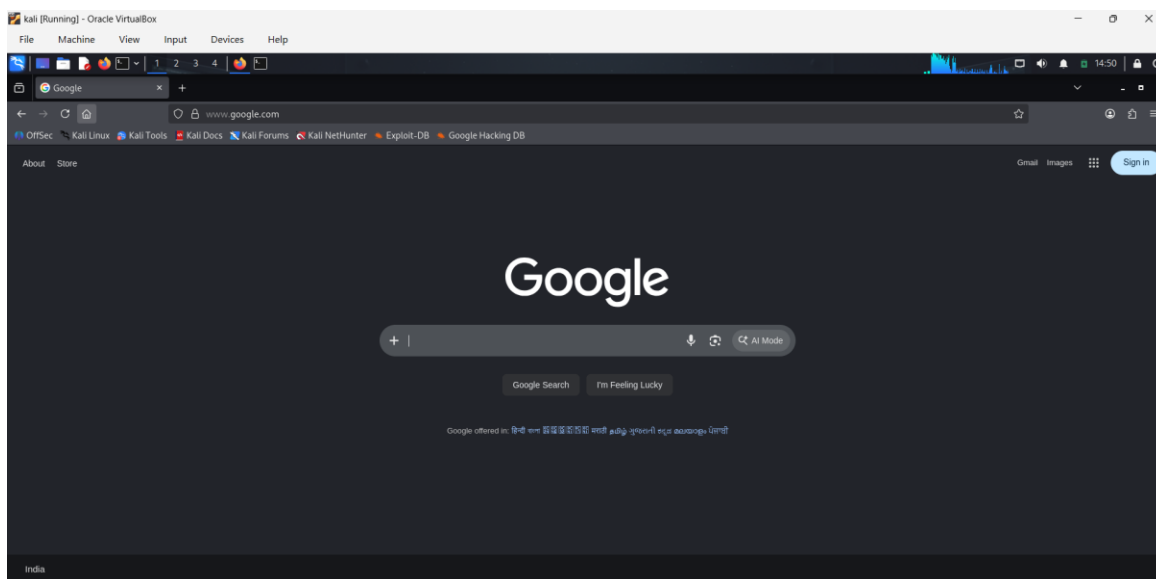


Figure 11: Successful HTTPS Page Load Confirming Port 443 Is Allowed

Testing SSH Connectivity to the Ubuntu Server

SSH access is tested from Kali Linux using the command:

```
ssh my-breach@192.168.230.177
```

The host key fingerprint is accepted, the password is entered, and the session connects successfully, showing the Ubuntu 24.04.4 LTS welcome banner. This confirms the Allow SSH rule correctly permits management access to the Ubuntu server on port 22.

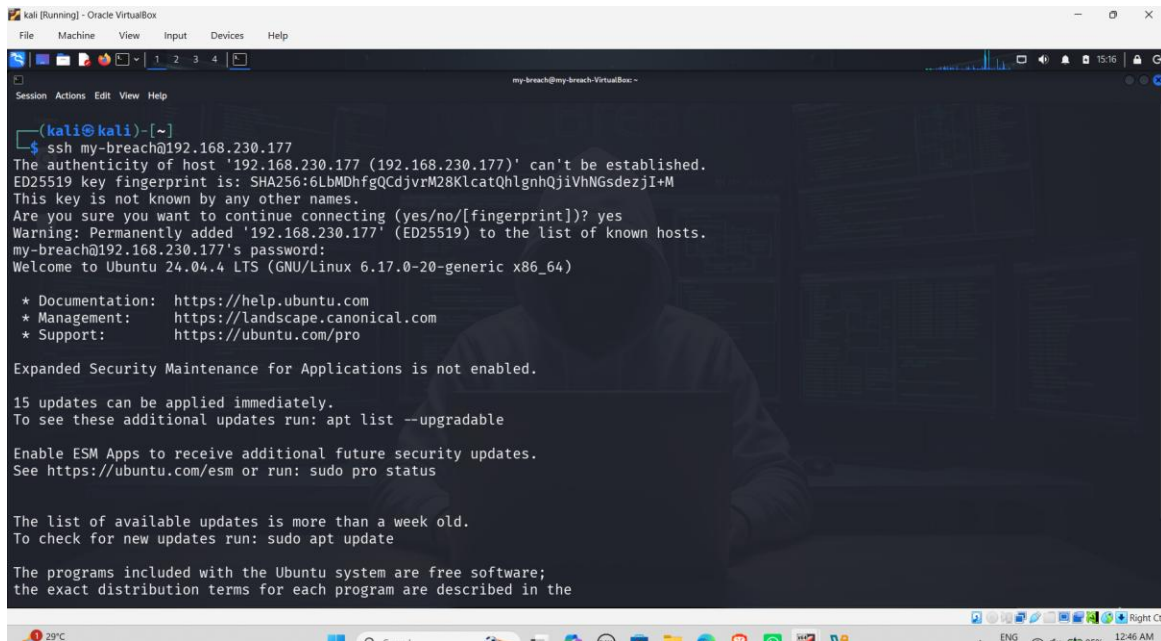


Figure 12: Successful SSH Login to the Ubuntu Server from Kali Linux

Validating ICMP, Telnet, and SMB Behaviour with Ping and Nmap

The Kali terminal is used to test the remaining protocols. A ping to 192.168.230.177 succeeds with replies received for all three packets, which shows that ICMP is still passing at this point even though a Block ICMP rule exists, since the default allow rule is still positioned ahead of it.

```
ping 192.168.230.177
nmap -Pn -p 23 192.168.230.177
nmap -Pn -p 445 192.168.230.177
```

The Nmap scans against port 23 and port 445 both report the ports as filtered rather than open, meaning traffic to Telnet and SMB is being dropped somewhere along the path, even though the block rules had not yet been moved above the allow-all rule.

```

kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
$ ping 192.168.230.177
PING 192.168.230.177 (192.168.230.177) 56(84) bytes of data:
64 bytes from 192.168.230.177: icmp_seq=1 ttl=64 time=5.61 ms
64 bytes from 192.168.230.177: icmp_seq=2 ttl=64 time=1.25 ms
64 bytes from 192.168.230.177: icmp_seq=3 ttl=64 time=1.35 ms
^C
--- 192.168.230.177 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 1.246/2.734/5.609/2.033 ms

kali@kali: ~
$ nmap -Pn -p 23 192.168.230.177
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-09 01:31 -0400
Nmap scan report for 192.168.230.177
Host is up (0.0015s latency).

PORT      STATE      SERVICE
23/tcp    filtered  telnet
MAC Address: 08:00:27:C5:74:C9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.93 seconds

kali@kali: ~
$ nmap -Pn -p 445 192.168.230.177
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-09 01:32 -0400
Nmap scan report for 192.168.230.177
Host is up (0.0011s latency).

PORT      STATE      SERVICE
445/tcp   filtered  microsoft-ds
MAC Address: 08:00:27:C5:74:C9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.88 seconds

```

Figure 13: ICMP, Telnet, and SMB Validation Using Ping and Nmap from Kali Linux

Reviewing Firewall Live Logs

The Firewall → Log Files → Live View page is opened to observe real-time traffic passing through OPNsense. The log shows WAN outbound entries for the firewall's own traffic and Homelab inbound entries labelled "anti-lockout rule" for HTTPS sessions to the OPNsense management interface at 192.168.230.2.

Reviewing the live log at this stage gives a baseline view of traffic activity before the rule ordering is changed to enforce the block rules properly.

Interface	Time	Protocol	Source	Destination	Action	Label
WAN	2026-07-09T11:05:14	UDP	192.168.10.3:22933	185.125.190.65:53	pass	let out anything from firewall host itself (force gw)
WAN	2026-07-09T11:05:14	UDP	192.168.10.3:52709	185.125.190.65:53	pass	let out anything from firewall host itself (force gw)
WAN	2026-07-09T11:05:07	UDP	192.168.10.3:123	192.46.211.253:123	pass	let out anything from firewall host itself (force gw)
WAN	2026-07-09T11:04:24	UDP	192.168.10.3:123	103.136.36.100:123	pass	let out anything from firewall host itself (force gw)
WAN	2026-07-09T11:04:08	UDP	192.168.10.3:4567	139.59.15.185:123	pass	let out anything from firewall host itself (force gw)
Homelab	2026-07-09T11:02:38	TCP	192.168.230.1:37529	192.168.230.2:443	pass	anti-lockout rule
Homelab	2026-07-09T11:02:38	TCP	192.168.230.1:6070	192.168.230.2:443	pass	anti-lockout rule
WAN	2026-07-09T11:02:12	UDP	192.168.10.3:123	103.136.36.100:123	pass	let out anything from firewall host itself (force gw)
WAN	2026-07-09T11:01:45	TCP	192.168.10.3:24815	91.189.91.97:80	pass	let out anything from firewall host itself (force gw)
WAN	2026-07-09T11:01:44	UDP	192.168.10.3:20933	185.125.190.65:53	pass	let out anything from firewall host itself (force gw)

Figure 14: Firewall Live View Showing Allowed and Passed Traffic

Reordering the Rule Set to Enforce Blocking

Because OPNsense applies the first matching rule, the Block telnet, Block SMB, and Block ICMP rules are moved above the default allow LAN rules on the Homelab interface. The updated list now shows Block telnet, Block SMB, and Block ICMP first, followed by Allow http, Allow https, Allow DNS, allow ssh, and finally the default allow LAN rules at the bottom.

The page confirms "The changes have been applied successfully," and a note at the bottom of the rule list explains that Homelab rules are evaluated on a first-match basis, so placing the block rules above the default allow rule is required for them to take effect.

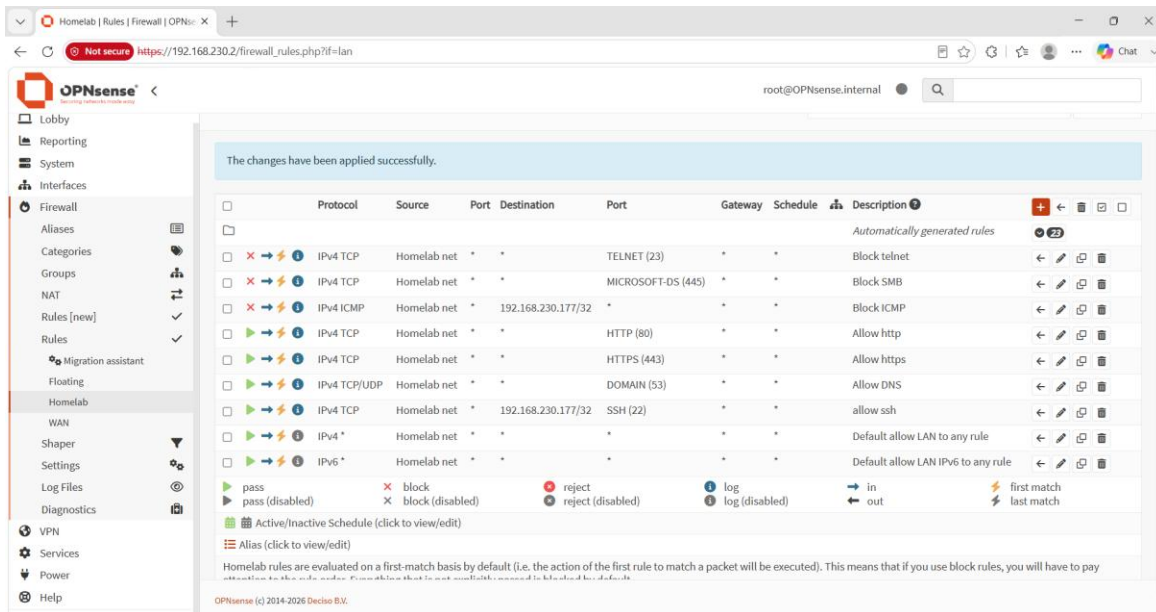


Figure 15: Final Rule Set with the Block Rules Positioned Above the Default Allow Rule

Disabling the Ubuntu UFW Firewall

To make sure that any blocking observed afterward is coming from OPNsense and not from the Ubuntu server's own firewall, UFW is disabled on the Ubuntu virtual machine with the command:

```
sudo ufw disable
```

The terminal confirms "Firewall stopped and disabled on system startup," removing UFW as a variable so the remaining tests validate OPNsense policy enforcement alone.

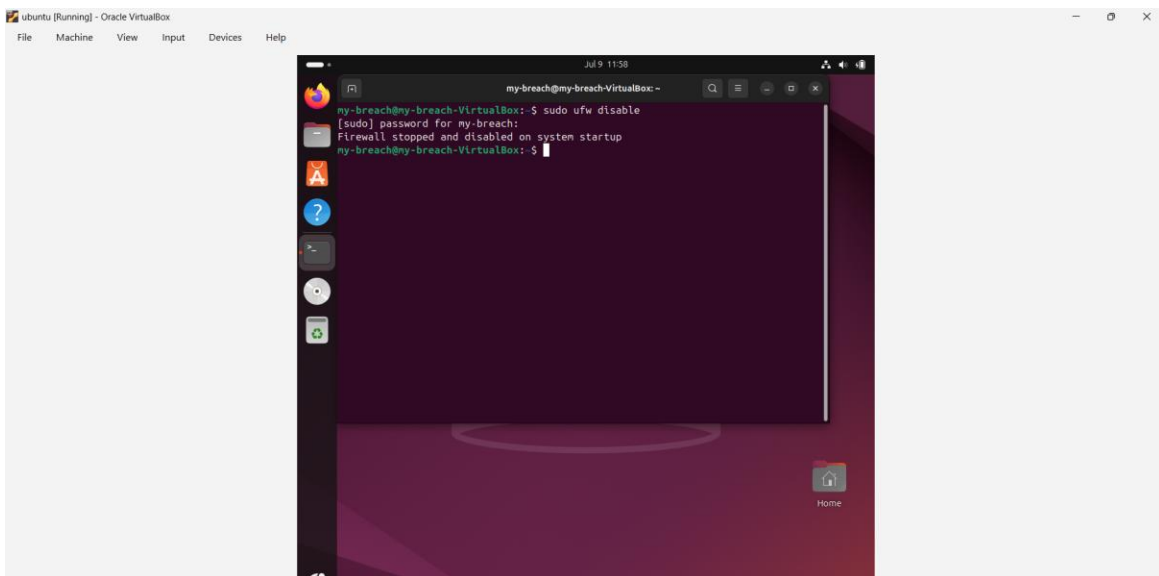


Figure 16: Disabling the Ubuntu UFW Firewall

Validating Telnet and SMB Blocking After Disabling UFW

With UFW disabled and the block rules now positioned above the allow-all rule, the Telnet and SMB tests are repeated from Kali Linux using:

```
nmap -Pn -p 23 192.168.230.177
nmap -Pn -p 445 192.168.230.177
```

This time both scans report the ports as closed rather than filtered, and the scans complete in under a second. With UFW disabled, the scans show both ports as **closed**, confirming that the configured firewall policy is being validated during testing.

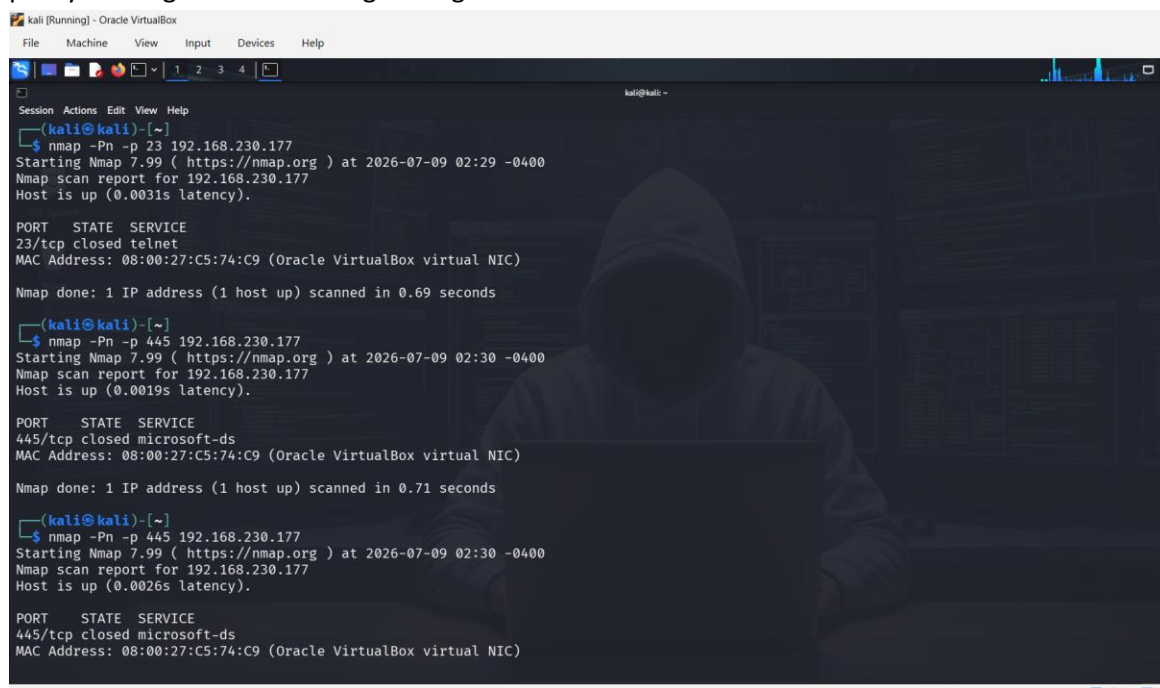


Figure 17: Telnet and SMB Ports Reported as Closed by OPNsense After Disabling UFW

Verifying Internet Connectivity to an External Host

To confirm that general internet connectivity still works after the rule changes, a ping is sent from Kali Linux to an external address:

```
ping 57.144.208.1
```

The external host replies to each ICMP echo request with round-trip times ranging from around 110 ms to nearly 300 ms, confirming that outbound connectivity from the Homelab network to the internet is still functioning normally after the firewall changes.

```
(kali@kali)-[~]
$ ping 57.144.208.1
PING 57.144.208.1 (57.144.208.1) 56(84) bytes of data.
64 bytes from 57.144.208.1: icmp_seq=1 ttl=62 time=116 ms
64 bytes from 57.144.208.1: icmp_seq=2 ttl=62 time=207 ms
64 bytes from 57.144.208.1: icmp_seq=3 ttl=62 time=192 ms
64 bytes from 57.144.208.1: icmp_seq=4 ttl=62 time=128 ms
64 bytes from 57.144.208.1: icmp_seq=5 ttl=62 time=161 ms
64 bytes from 57.144.208.1: icmp_seq=6 ttl=62 time=172 ms
64 bytes from 57.144.208.1: icmp_seq=7 ttl=62 time=287 ms
64 bytes from 57.144.208.1: icmp_seq=8 ttl=62 time=263 ms
64 bytes from 57.144.208.1: icmp_seq=9 ttl=62 time=249 ms
64 bytes from 57.144.208.1: icmp_seq=10 ttl=62 time=294 ms
64 bytes from 57.144.208.1: icmp_seq=11 ttl=62 time=311 ms
64 bytes from 57.144.208.1: icmp_seq=12 ttl=62 time=130 ms
64 bytes from 57.144.208.1: icmp_seq=13 ttl=62 time=110 ms
64 bytes from 57.144.208.1: icmp_seq=14 ttl=62 time=198 ms
64 bytes from 57.144.208.1: icmp_seq=15 ttl=62 time=182 ms
64 bytes from 57.144.208.1: icmp_seq=16 ttl=62 time=188 ms
64 bytes from 57.144.208.1: icmp_seq=17 ttl=62 time=173 ms
64 bytes from 57.144.208.1: icmp_seq=18 ttl=62 time=143 ms
```

Figure 18: ICMP Echo Requests Successfully Reaching an External Host

Observing Allowed Traffic in the Firewall Live View

The Firewall → Log Files → Live View page is checked again, this time to observe entries for the traffic that was just generated. Passed connections on the Homelab interface for HTTP, HTTPS, DNS, and SSH appear with the action set to pass, matching the rules configured earlier in the lab.

Seeing these entries confirms that the allow rules are being logged and matched correctly even after the block rules were moved above the default allow rule.

Firewall: Log Files: Live View							
Filters				Templates		Options	
Quick search (all fields)...				Choose template		☒ Auto-refresh	
action contains Search...						☐ Lookup hostnames	
Active filters:						☐ Select any of given criteria (or)	
						25 Table size	
						10000 History size	
Interface	Time	Protocol	Source	Destination	Action	Label	
WAN	Out	2026-07-09T12:06:33	ICMP	192.168.10.3	57.144.208.1	pass	let out anything from firewall host itself (force gw)
WAN	Out	2026-07-09T12:06:32	UDP	192.168.10.3:123	192.46.211.253:123	pass	let out anything from firewall host itself (force gw)
WAN	Out	2026-07-09T12:06:21	ICMP	192.168.10.3	57.144.208.1	pass	let out anything from firewall host itself (force gw)
WAN	Out	2026-07-09T12:05:56	UDP	192.168.10.3:68	192.168.10.2:67	pass	allow DHCP client on WAN
WAN	Out	2026-07-09T12:05:50	UDP	192.168.10.3:123	103.136.36.100:123	pass	let out anything from firewall host itself (force gw)
WAN	Out	2026-07-09T12:05:00	UDP	192.168.10.3:28449	3.7.223.15:123	pass	let out anything from firewall host itself (force gw)
WAN	Out	2026-07-09T12:04:44	ICMP	192.168.10.3	8.8.8.8	pass	let out anything from firewall host itself (force gw)
WAN	Out	2026-07-09T12:04:42	TCP	192.168.10.3:60827	185.125.190.100:80	pass	let out anything from firewall host itself (force gw)
Homelab	In	2026-07-09T12:04:42	TCP	192.168.230.177:44792	185.125.190.100:80	pass	Allow http
WAN	Out	2026-07-09T12:04:42	UDP	192.168.10.3:25864	185.125.190.65:53	pass	let out anything from firewall host itself (force gw)

Figure 19: Firewall Live View Showing Allowed Network Traffic

Enabling Logging on the Block ICMP Rule

To capture blocked ICMP attempts in the live log, the Block ICMP rule is edited and the "Log packets that are handled by this rule" checkbox is enabled. The rule still applies to the Homelab interface, Protocol ICMP, Source Homelab network, and Destination 192.168.230.177, with the Description left as "Block ICMP".

Turning logging on for this rule makes it possible to see blocked ping attempts appear in the Live View in the next steps.

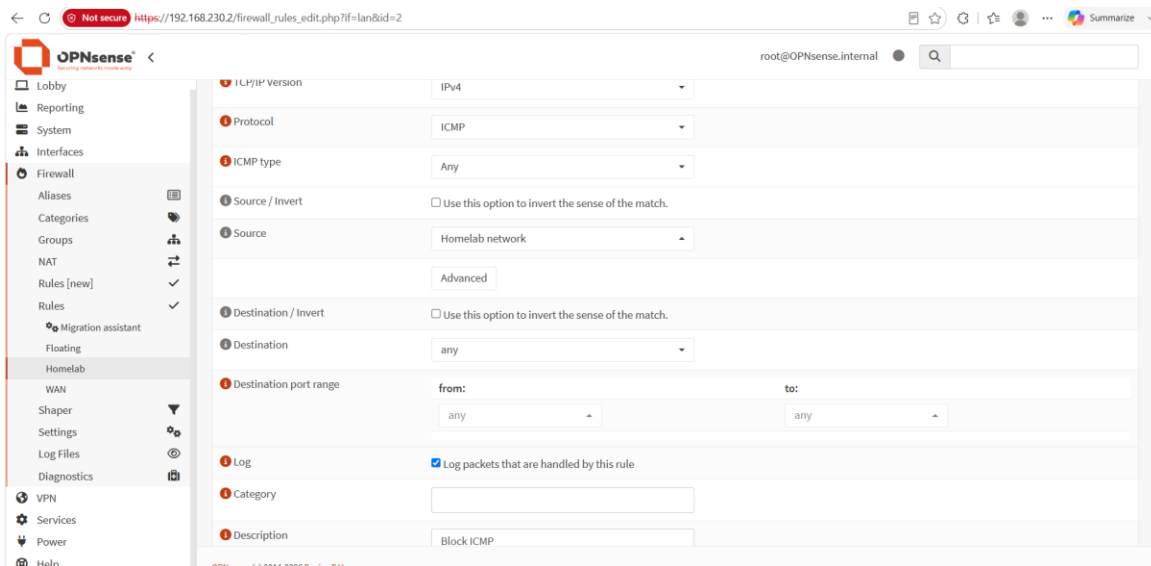


Figure 20: Enabling Logging on the Block ICMP Rule

Validating Internal and External ICMP Behaviour

The ping to 192.168.230.177 is successful with 0% packet loss, confirming connectivity to the Ubuntu server. A second ping to 57.144.208.1 receives 100% packet loss, showing that no icmp replies are received from the external destination during this test.

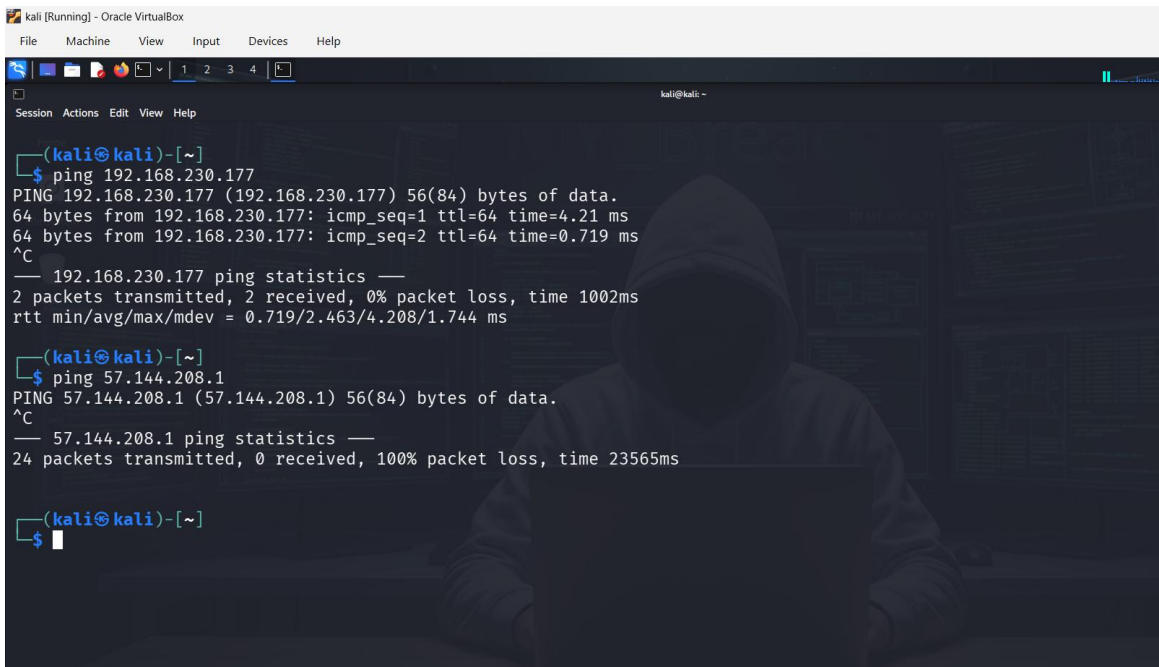


Figure 21: Successful Internal ICMP Connectivity and Blocked External ICMP Test

Observing Blocked ICMP Traffic in the Live View

Returning to Firewall → Log Files → Live View, the blocked ping attempts now appear in the log with the action set to block and the label "Block ICMP," showing the source as the Kali Linux host and the destination as 192.168.230.177.

This confirms end-to-end that the Block ICMP rule is not only preventing ping traffic from reaching the Ubuntu server but is also being logged correctly for monitoring purposes.

Firewall: Log Files: Live View

Filters: Quick search (all fields)... [Refresh] [Save]

Templates: [Choose template] [Delete]

Options: ☒ Auto-refresh, ☐ Lookup hostnames, ☐ Select any of given criteria (or)

Table size: 25, History size: 10000

Interface		Time	Protocol	Source	Destination	Action	Label	
Homelab	In	2026-07-09T12:09:38	ICMP	192.168.230.99	57.144.208.1	block	Block ICMP	1
Homelab	In	2026-07-09T12:09:37	ICMP	192.168.230.99	57.144.208.1	block	Block ICMP	1
Homelab	In	2026-07-09T12:09:36	ICMP	192.168.230.99	57.144.208.1	block	Block ICMP	1
Homelab	In	2026-07-09T12:09:35	ICMP	192.168.230.99	57.144.208.1	block	Block ICMP	1
Homelab	In	2026-07-09T12:09:34	ICMP	192.168.230.99	57.144.208.1	block	Block ICMP	1
Homelab	In	2026-07-09T12:09:33	ICMP	192.168.230.99	57.144.208.1	block	Block ICMP	1
Homelab	In	2026-07-09T12:09:32	ICMP	192.168.230.99	57.144.208.1	block	Block ICMP	1
Homelab	In	2026-07-09T12:09:31	ICMP	192.168.230.99	57.144.208.1	block	Block ICMP	1
Homelab	In	2026-07-09T12:09:30	ICMP	192.168.230.99	57.144.208.1	block	Block ICMP	1
Homelab	In	2026-07-09T12:09:29	ICMP	192.168.230.99	57.144.208.1	block	Block ICMP	1

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Figure 22: Blocked ICMP Traffic Recorded in the Firewall Live View

Final Firewall Policy on the Homelab Interface

The Homelab rule list is reviewed one final time to document the completed policy. The Block telnet, Block SMB, and Block ICMP rules sit above the Allow http, Allow https, Allow DNS, and allow ssh rules, with the default allow LAN rules at the bottom, matching the order required for the block rules to take effect.

This final configuration reflects everything validated throughout the lab: HTTP, HTTPS, DNS, and SSH traffic is permitted, while Telnet, SMB, and ICMP to the Ubuntu server are blocked and logged.

Firewall: Rules: Homelab

Select category

Inspect

☐	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description					
	Automatically generated rules										23		
☐	IPv4 TCP	Homelab net	*	*	TELNET (23)	*	*	Block telnet					
☐	IPv4 TCP	Homelab net	*	*	MICROSOFT-DS (445)	*	*	Block SMB					
☐	IPv4 ICMP	Homelab net	*	*	*	*	*	Block ICMP					
☐	IPv4 TCP	Homelab net	*	*	HTTP (80)	*	*	Allow http					
☐	IPv4 TCP	Homelab net	*	*	HTTPS (443)	*	*	Allow https					
☐	IPv4 TCP/UDP	Homelab net	*	*	DOMAIN (53)	*	*	Allow DNS					
☐	IPv4 TCP	Homelab net	*	192.168.230.177/32	SSH (22)	*	*	allow ssh					
☐	IPv4 *	Homelab net	*	*	*	*	*	Default allow LAN to any rule					
☐	IPv6 *	Homelab net	*	*	*	*	*	Default allow LAN IPv6 to any rule					
	pass	block			reject			log			in		
	pass (disabled)	block (disabled)			reject (disabled)			log (disabled)			out		
	Active/Inactive Schedule (click to view/edit)												
	Alias (click to view/edit)												
Homelab rules are evaluated on a first-match basis by default (i.e. the action of the first rule to match a packet will be executed). This means that if you use block rules, you will have to pay attention to the rule order. Everything that is not explicitly passed is blocked by default.													

Figure 23: Final OPNsense Firewall Policy Configured on the Homelab Interface

Conclusion

The firewall rules for HTTP, HTTPS, DNS, SSH, Telnet, SMB, and ICMP were configured successfully on the Homelab interface in OPNsense. Allowed traffic for HTTP, HTTPS, DNS, and SSH was verified from Kali Linux, and blocked traffic for Telnet, SMB, and ICMP was confirmed once the block rules were positioned above the default allow rule.

Testing was carried out using ping, nslookup, ssh, and nmap from Kali Linux, and the Ubuntu server's own UFW firewall was disabled so that the results reflected OPNsense enforcement alone. The Firewall Live View was used throughout to confirm that both allowed and blocked traffic were being logged correctly.

The testing confirms that the configured OPNsense firewall rules operate as expected. Allowed services remain accessible, while the configured blocked services are successfully restricted and recorded in the firewall logs.